Preface

This is the second course in a two-quarter long sequence covering functional analysis, Hilbert space theory, and partial differential equations. Most of the material is covered in Ward Cheney's *Analysis for Applied Mathematics*, Nicholas Young's *An Introduction to Hilbert Space*, and Lawrence C. Evans's *Partial Differential Equations*.

Contents

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Chapter 1
Preliminaries

For the remainder of this chapter: let $U \subset \mathbb{R}^n$ be an open set. An element $x \in \mathbb{R}^n$ will always be of the form $(x_1, \ldots, x_n)$. The function $u : U \to \mathbb{R}$ will always denote the unknown function being solved for in a differential equation. We denote $u := u(x) = u(x_1, \ldots, x_n)$.

§ 1.1. Basic Definitions

Definition 1.1.1. The *unit vector in* $\mathbb{R}^n$ is denoted by e^i, where it is 1 is the i^{th} slot and 0 otherwise.

Definition 1.1.2. The *partial derivative* of u with respect to x_i is:

$$u_{x_i} := \frac{\partial u}{\partial x_i} := \lim_{h \to 0} \frac{u(x + he^i) - u(x)}{h}.$$

Definition 1.1.3. The *gradient* of u is the vector:

$$Du := \nabla u := (u_{x_1}, \ldots, u_{x_n}).$$

Definition 1.1.4. The *Hessian matrix* of u is the 2 dimensional array:

$$D^2 u = \begin{pmatrix} u_{x_1 x_1} & \cdots & u_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ u_{x_n x_1} & \cdots & u_{x_n x_n} \end{pmatrix}$$

Definition 1.1.5. The *Laplacian* of u is:

$$\Delta u := \nabla^2 u := \mathrm{Tr}(D^2 u) = \sum_{i=1}^{n} u_{x_i x_i}.$$

Definition 1.1.6.

(a) A vector of the form $\alpha := (\alpha_1, \ldots, \alpha_n)$, where $\alpha_i \in \mathbb{Z}_{\geqslant 0}$, is called a *multiindex of order* $|\alpha|$, where:

$$|\alpha| = \alpha_1 + \ldots + \alpha_n.$$

(b) Give a multiindex α, we define:

$$D^\alpha u(x) := \frac{\partial^{|\alpha|} u(x)}{\partial^{\alpha_1} x_1 \ldots \partial^{\alpha_n} x_n}.$$

Definition 1.1.7. A *partial differential equation* (PDE for short) is an equation involving an unknown function u along with one or more of its partial derivatives. A partial differential equation is *linear* if we can write it as:

$$\sum_{|a|\leqslant K} a_\alpha(x)D^\alpha u(x) = f(x).$$

If $f(x) = 0$, then this PDE is called *homogenous*.

§ 1.2. Multivariable Calculus

Definition 1.2.1. Let $U \subset \mathbb{R}^n$ and $f \in L^1(\mathbb{R},\mu)$. The *average of f* over U is:

$$⨍ f\,d\mu := \frac{1}{\mu(U)}\int_U f\,d\mu.$$

Theorem 1.2.1 (Fubini). *If $u \in L^1(U)$, then:*

$$\int_U u\,dx = \underbrace{\int \dots \int}_{n\text{-times}} u(x_1,\dots,x_n)dx_1\dots dx_n.$$

Definition 1.2.2. The *boundary* of U is denoted ∂U.

Definition 1.2.3. We say the boundary ∂U is *C^k* if in a neighborhood of any point on the boundary, the boundary is the graph of a C^k function (possibly after the coordinate system has been rotated).

Definition 1.2.4. If ∂U is C^1, then along ∂U is defined the *outward pointing unit normal vector field*, which we denote by:

$$\vec{\nu} = (\nu^1,\dots,\nu^n).$$

Definition 1.2.5. Let $u \in C^1(\overline{U})$. The *outward normal derivative* of u is:

$$\frac{\partial u}{\partial \nu} := Du\cdot\vec{\nu}.$$

Definition 1.2.6. Let $\vec{u} = (u^1,\dots,u^n)$ be a vector field. The *flux of u across ∂U* is:

$$\int_{\partial U} \vec{u}\cdot\vec{\nu}\,dS.$$

Theorem 1.2.2 (Gauss-Green).

(a) Suppose $u \in C^1(\overline{U})$. For any $i = 1,\dots,n$, we have:

$$\int_U u_{x_i}\,dx = \int_{\partial U} u\nu^i\,dS$$

(b) *(Divergence Theorem) Let* $\vec{v} = (v^1, \ldots, v^n)$*, where* $v^i \in C^1(\overline{U})$ *for each* $i = 1, \ldots, n$*. Then:*

$$\int_U \operatorname{div} \vec{v} dx = \int_{\partial U} \vec{v} \cdot \vec{\nu} dS.$$

Theorem 1.2.3 (Integration by Parts). *Let* $u, v \in C^1(\overline{U})$ *and let* ∂U *be* C^1*. Then:*

$$\int_U u_{x_i} v dx = -\int_U u v_{x_i} dx + \int_{\partial U} u v \nu^i dS.$$

Theorem 1.2.4 (Green's Formulas). *Let* $u, v \in C^1(\overline{U})$ *and let* ∂U *be* C^1*. Then:*

(a) $\displaystyle\int_U \Delta u dx = \int_{\partial U} \frac{\partial u}{\partial \nu} dS,$

(b) $\displaystyle\int_U Dv \cdot Du dx = -\int_U (\Delta v) u \, dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u \, dS,$

(c) $\displaystyle\int_U u\Delta v - v\Delta u \, dx = \int_{\partial U} u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} dS.$

§ 1.3. Polar Coordinates in $\mathbb{R}^n$

Remark. In $\mathbb{R}^n$, for any $x \neq 0$, we can write:

$$x = |x| \cdot \frac{x}{|x|} := r\omega,$$

where $r \in (0, \infty)$ and $\omega \in S^{n-1} = \partial B(0,1)$.

Theorem 1.3.1. *There exists a measure* $d\omega$ *on* S^{n-1} *such that:*

$$\int_{\mathbb{R}^n} u dx = \int_0^\infty \left(\int_{S^{n-1}} u(r\omega) r^{n-1} d\omega \right) dr.$$

In other words, $dx = r^{n-1} d\omega dr$*.*

Corollary 1.3.2. *For any* $x_0 \in \mathbb{R}^n$ *and* $a > 0$*, we have:*

$$\int_{B(x_0,a)} u dx = \int_{B(0,a)} u(x_0 + y) dy = \int_0^a \left(\int_{S^{n-1}} u(x_0 + r\omega) r^{n-1} d\omega \right) dr.$$

Corollary 1.3.3. *For any* $x_0 \in \mathbb{R}^n$ *and* $r > 0$*, we have:*

$$\frac{d}{dr} \int_{B(x_0,r)} u dx = \int_{\partial B(x_0,r)} u dS.$$

§ 1.4. Smoothing

Definition 1.4.1.

(a) Let $U, V \subset \mathbb{R}^n$. We say U is *compactly contained* in V, and write $U \subset\subset V$, if $V \subset \overline{V} \subset U$ and $\overline{V}$ is compact.

(b) If $f|_V \in L^p(V)$ for all $V \subset\subset U$, then we say f is *locally integrable on U*. The set of all such functions is denoted $L^p_{\text{loc}}(U)$.

Definition 1.4.2.

(a) Let $x \in \mathbb{R}^n$. The *standard mollifier* is the function:

$$\eta(x) := \begin{cases} \frac{-C}{1-|x|^2}, & |x| < 1 \\ 0, & |x| \geqslant 1 \end{cases},$$

where C is chosen so that $\int_{\mathbb{R}^n} \eta(x) dx = 1$.

(b) For all $\epsilon > 0$, define $\eta_\epsilon(x) := \frac{1}{\epsilon^n} \eta\left(\frac{x}{\epsilon}\right)$.

Remark. The standard mollifier is a radial function.

Proposition 1.4.1.

(a) $\eta_\epsilon \in C_c^\infty(\mathbb{R}^n)$;

(b) $\operatorname{supp} \eta_\epsilon = B(0, \epsilon)$;

(c) $\int_{\mathbb{R}^n} \eta_\epsilon(x)\, dx = 1$.

Theorem 1.4.2. *Let $U \subset \mathbb{R}^n$ be open and $f \in L^1_{\text{loc}}(U)$. Define $U_\epsilon := \{y \in U : \epsilon < \operatorname{dist}(y, \partial U)\}$ and $f^\epsilon(x) := (f * \eta_\epsilon)(x)$. Then:*

(a) $f^\epsilon \in C^\infty(U_\epsilon)$;

(b) If $f \in C(U)$, then $f^\epsilon \to f$ uniformly on compact subsets of U.

Chapter 2
Solutions to Linear Partial Differential Equations

§ 2.1. The Transport Equation

Definition 2.1.1. The *Transport Equation* is:

$$\begin{cases} u_t + b \cdot Du = 0, & \text{in } \mathbb{R}^n \times (0, \infty) \end{cases},$$

where b is a fixed vector in $\mathbb{R}^n$ and $u : \mathbb{R}^n \times (0, \infty) \to \mathbb{R}$ is the unknown function.

Theorem 2.1.1. *Let $g \in C^1(\mathbb{R}^n)$. Then $u(x,t) := g(x - tb)$ is a solution to the following initial value problem:*

$$\begin{cases} u_t + Du \cdot b = 0, & \text{in } \mathbb{R}^n \times (0, \infty) \\ u(x,0) = g(x), & \text{on } \mathbb{R}^n \times \{0\} \end{cases}.$$

Proof. As b is an arbitrary vector in $\mathbb{R}^n$, the trick is to paramtrize the line through (x,t) with direction $(b, 1)$. Define $z(s) := u(x + sb, t + s)$. Then:

$$\begin{aligned} \dot{z}(s) &= \frac{d}{ds} u(x + sb, t + s) \\ &= \left(\sum_{j=1}^{n} u_{x_j}(x + sb, t + s) b_j \right) + u_t \\ &= Du \cdot b + u_t \\ &= 0. \end{aligned}$$

Since z is constant, we have for all $(x,t) \in \mathbb{R}^n \times (0, \infty)$:

$$\begin{aligned} u(x,t) &= z(0) \\ &= z(-t) \\ &= u(x - tb, 0) \\ &= g(x - tb). \end{aligned}$$

■

Theorem 2.1.2. *Let $g \in C^1(\mathbb{R}^n)$. Then $u(x,t) := g(x - tb) + \int_0^t f(x + (s+t)b, s)\, ds$ is a solution to the following non-homogenous initial value problem:*

$$\begin{cases} u_t + Du \cdot b = f, & \text{in } \mathbb{R}^n \times (0, \infty) \\ u(x,0) = g(x), & \text{on } \mathbb{R}^n \times \{0\} \end{cases}.$$

Proof. Similarly define $z(s) := u(x+sb, s+t)$. Then $\dot{z}(s) = Du \cdot b = f(x+sb, s+t)$. By the Fundamental Theorem of Calculus:

$$\begin{aligned} u(x,t) &= z(0) \\ &= z(-t) + \int_{-t}^{0} f(x+sb, s+t)\, ds \\ &= g(x-tb) + \int_{-t}^{0} f(x+sb, s+t)\, ds. \end{aligned}$$

Making the substitution $s \mapsto t+s$, gives the desired result. ■

§ 2.2. Laplace's Equation

Definition 2.2.1. *Laplace's equation* is:

$$\Delta u = 0.$$

A C^2 function satisfying Laplace's equation is called a *harmonic function*.

I. The Fundamental Solution

Proposition 2.2.1. *Laplace's equation is rotation invariant. Specifically, let $A = (a_{ij})$ denote an $n \times n$ orthogonal matrix. For $x \in \mathbb{R}^n$, let $y = Ax$; that is:*

$$Ax = y = (y_1, \ldots, y_n), \; y_i = \sum_{j=1}^{n} a_{ij} x_i$$

Assume $u \in C^2(\mathbb{R}^n)$ satisfies $\Delta u = 0$. Then define

$$v(x) = u(Ax) = u(y).$$

We have that $\Delta v = 0$.

Proof. Note that $\frac{\partial y_k}{\partial x_i} = a_{ki}$. Hence:

$$\begin{aligned} v_{x_i}(x) &= \frac{\partial}{\partial x_i} u(y) \\ &= \sum_{k=1}^{n} u_{x_k}(y) \frac{\partial y_k}{\partial x_i} \\ &= \sum_{k=1}^{n} u_{x_k}(y) a_{ki}. \end{aligned}$$

Taking a second partial derivative with respect to x_i gives:

$$\begin{aligned} v_{x_i x_i}(x) &= \frac{\partial}{\partial x_i} \sum_{k=1}^{n} u_{x_k}(y) a_{ki} \\ &= \sum_{k=1}^{n} \sum_{j=1}^{n} u_{x_k x_j}(y) \frac{\partial y_j}{\partial x_i} a_{ki} \\ &= \sum_{j=1}^{n} \sum_{k=1}^{n} a_{ji} a_{ki}\, u_{x_j x_k}. \end{aligned}$$

Summing from $i = 1, \ldots, n$ finally yields:

$$\begin{aligned} \Delta v(x) &= \sum_{i=1}^{n} v_{x_i x_i}(x) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} a_{ji} a_{ki}\, u_{x_j x_k}(y) \\ &= \sum_{j=1}^{n} \sum_{k=1}^{n} \left(\sum_{i=1}^{n} a_{ji} a_{ki} \right) u_{x_j x_k}(y) \\ &= \sum_{j=1}^{n} \sum_{k=1}^{n} \delta_{jk}\, U_{x_j x_k}(y) \\ &= \sum_{k=1}^{n} u_{x_k x_k} \\ &= \Delta u(y) \\ &= 0 \end{aligned}$$

■

Example 2.2.1. Since solutions to Laplace's equation are rotation invariant, we are interested in finding *radial solutions* to the equation; i.e., functions of $r = |x|$. Let u be a solution of Laplace's equation and suppose further we have $u(x) = v(r)$ Then $u_{x_i} = v'(r) r_{x_i}$ and $u_{x_i x_i} = v''(r) r_{x_i}^2 + v'(r) r_{x_i x_i}$. Using the fact that:

$$\begin{aligned} \frac{\partial}{\partial x_i} r^2 &= \frac{\partial}{\partial x_i} \sum_{i=1}^{n} x_i^2 \\ &= 2 r r_{x_i}, \end{aligned}$$

we get that $r_{x_i} = \frac{x_i}{r}$. A similar calculation yields $r_{x_i x_i} = \frac{1}{r} - \frac{x_i^2}{r^3}$. Now since u is a solution to Laplace's equation, we have that $\Delta u = 0$. We can write this as:

$$\sum_{i=1}^{n} u_{x_i x_i} = 0.$$

Substituting for our earlier calculation gives:

$$\sum_{i=1}^{n} v''(r) \frac{x_i^2}{r^2} + v'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right) = 0.$$

Hence:

$$v''(r) + v'(r)\frac{(n-1)}{r} = 0.$$

For $n = 2$, the solution to the above ODE is $v(r) = c_1 \log r + c_2$. For $n > 2$, the solution to the above ODE is $v(r) = c_1 \frac{1}{r^{n-1}} + c_2$.

Definition 2.2.2. The *fundamental solution* to Laplace's equation is:

$$\Phi(x) = \begin{cases} c_2 \log \frac{1}{|x|}, & n = 2 \\ c_n |x|^{2-n}, & n > 2, \end{cases}$$

where $c_2 = \frac{1}{2\pi}$ and $c_n = \frac{1}{n(n-2)\alpha(n)}$, where $\alpha(n)$ denotes the volume of $B(0,1)$ in $\mathbb{R}^n$.

Remark. The fundamental solution to Laplace's equation admits a singularity at $x = 0$.

Remark. If u and v are both harmonic, then $au + bv$ is harmonic for any $a, b \in \mathbb{R}$ since the Laplacian is a linear differential operator. Moreover, if u is harmonic then $u(x - x_0)$ is also harmonic when treated as a variable with respect to x. These reasons combined lead to the following conclusion: if $y^1, \ldots, y^k \in \mathbb{R}^n$ and $f_1, \ldots, f_k \in \mathbb{R}$, the function:

$$\sum_{i=1}^{k} f_i \Phi(x - y^i)$$

is also harmonic! A natural question to as now is whether:

$$\int_{\mathbb{R}^n} f(y)\Phi(x-y)dy$$

is harmonic.

Theorem 2.2.2. *Let $f \in C_c^2(\mathbb{R})$. Define $u(x) = \int_{\mathbb{R}^n} f(y)\Phi(x-y)dy$. Then $u \in C^2(\mathbb{R}^n)$ and $-\Delta u = f$.*

Proof. Taking the difference quotient of $u(x)$ gives the following equality:

$$\frac{u(x+he^i) - u(x)}{h} = \int_{\mathbb{R}^n} \frac{f(x+he^i - y) - f(x-y)}{h}\Phi(y)dy.$$

Applying the Dominated Convergence Theorem gives:

$$u_{x_i} = \int_{\mathbb{R}^n} f_{x_i}(x-y)\Phi(y)dy.$$

Taking the difference quotient of both sides and applying the Dominated Convergence Theorem again yields:

$$u_{x_i x_i} = \int_{\mathbb{R}^n} f_{x_i x_i}(x-y)\Phi(y)dy.$$

Thus:

$$\begin{aligned}\Delta u &= \int_{\mathbb{R}^n} \Delta_x f(x-y)\Phi(y)dy \\ &= \int_{\mathbb{R}^n} (-1)^2 \Delta_y f(x-y)\Phi(y)dy.\end{aligned}$$

Since f is a function with compact support, let $U := B^o(0,R)$ be large enough such that $x \in U$ and $\operatorname{supp} f(x-y) \subset U$. Our integral becomes:

$$\begin{aligned}\Delta u &= \int_{\mathbb{R}^n} \Delta_y f(x-y)\Phi(y)dy \\ &= \int_U \Delta_y f(x-y)\Phi(y)dy.\end{aligned}$$

Recall that our fundamental solution Φ admits a discontinuity at the origin. For an arbitrary $\epsilon > 0$, we have:

$$\Delta u = \int_{B(0,\epsilon)} \Delta_y f(x-y)\Phi(y)dy + \int_{U\setminus B(0,\epsilon)} \Delta_y f(x-y)\Phi(y)dy$$

Since $\Delta_y f(x-y)\Phi(y)$ does not admit a discontinuity on $U \setminus B(0,\epsilon)$, we can apply Green's identities. Taking advantage of the fact that $\Delta_y \Phi(y) = 0$, we have:

$$\begin{aligned}\int_{U\setminus B(0,\epsilon)} \Delta_y f(x-y)\Phi(y)dy &= \int_{U\setminus B(0,\epsilon)} \Delta_y f(x-y)\Phi(y) - f(x-y)\Delta_y \Phi(y)dy \\ &= \int_{\partial(U\setminus B(0,\epsilon))} \frac{\partial f}{\partial \nu}(x-y)\Phi(y) - f(x-y)\frac{\partial \Phi}{\partial \nu}(y)dS(y).\end{aligned}$$

Integrating on $\partial(U \setminus B(0,\epsilon)$ is equivalent to:

$$\int_{\partial U} \frac{\partial f}{\partial \nu}(x-y)\Phi(y) - f(x-y)\frac{\partial \Phi}{\partial \nu}(y)dS(y) - \int_{\partial B(0,\epsilon)} \frac{\partial f}{\partial \nu}(x-y)\Phi(y) - f(x-y)\frac{\partial \Phi}{\partial \nu}(y)dS(y).$$

The outward normal derivative of f is zero, and since f is of compact support it is zero along the boundary of U. The left-most integral is equal to zero, resulting in:

$$\Delta u = \underbrace{\int_{B(0,\epsilon)} \Delta_y f(x-y)\Phi(y)dy}_{:=I_\epsilon} - \underbrace{\int_{\partial B(0,\epsilon)} \frac{\partial f}{\partial \nu}(x-y)\Phi(y)\, dS(y)}_{:=K_\epsilon} + \underbrace{\int_{\partial B(0,\epsilon)} f(x-y)\frac{\partial \Phi}{\partial \nu}(y)\, dS(y)}_{:=L_\epsilon}$$

We will individually examine I_ϵ, K_ϵ, and L_ϵ as $\epsilon \to \infty$. We first start with I_ϵ. Observe that:

$$\begin{aligned}
|I_\epsilon| &\leqslant \int_{B(0,\epsilon)} |\Delta_y f(x-y)| \Phi(y)\, dy \\
&\leqslant C \int_{B(0,\epsilon)} \Phi(y)\, dy \\
&\leqslant C \int_{B(0,\epsilon)} |x|^{2-n}\, dy \\
&= C \int_0^\epsilon \int_{S^{n-1}} r^{2-n} r^{n-1}\, dr d\omega \\
&= C \int_0^\epsilon r \left(\int_{S^{n-1}} d\omega \right) dr \\
&\leqslant C \int_0^\epsilon r\, dr \\
&\leqslant C\epsilon^2.
\end{aligned}$$

Thus $I_\epsilon \to 0$. We next examine K_ϵ. Observe that:

$$\begin{aligned}
|K_\epsilon| &\leqslant \int_{\partial B(0,\epsilon)} \left| \frac{\partial f}{\partial \nu}(x-y) \right| \Phi(y)\, dy \\
&\leqslant C \int_{\partial B(0,\epsilon)} \Phi(y)\, dS(y) \\
&\leqslant C \int_{\partial B(0,\epsilon)} |y|^{2-n}\, dS(y) \\
&= C\epsilon^{2-n} \int_{\partial B(0,\epsilon)} dS(y) \\
&\leqslant C\epsilon^{2-n}\epsilon^{n-1} \\
&= C\epsilon.
\end{aligned}$$

Thus $K_\epsilon \to 0$. The last and most difficult integral to examine is L_ϵ. We must first compute $\frac{\partial \Phi}{\partial \nu}$. Computing a partial of Φ yields:

$$\begin{aligned}
\frac{\partial}{\partial y_i}\Phi(y) &= c_n \frac{\partial}{\partial y_i} |y|^{2-n} \\
&= c_n(2-n)|y|^{1-n} \frac{\partial}{\partial y_i}|y| \\
&= -c_n(n-2)|y|^{1-n} \frac{1}{2} \left(\sum_{j=1}^n y_j^2 \right)^{-\frac{1}{2}} 2y_i \\
&= -c_n(n-2)|y|^{1-n} \frac{y_i}{|y|} \\
&= -c_n(n-2)|y|^{-n} y_i.
\end{aligned}$$

Hence $D\Phi(y) = -c_n(n-2)\frac{1}{2}|y|^{2-n}y$. Since $y \in \partial B(0,\epsilon)$, it already points in the direction as the outward normal;i.e., $\nu = \frac{y}{|y|}$. With this:

$$\begin{aligned}
\frac{\partial \Phi}{\partial \nu}(y) &= D\Phi(y)\cdot \nu \\
&= -c_n(n-2)|y|^{-n}y \cdot \frac{y}{|y|} \\
&= -c_n(n-2)|y|^{-n-1}(y\cdot y) \\
&= -c_n(n-2)|y|^{-n-1}|y|^2 \\
&= -c_n(n-2)|y|^{1-n}.
\end{aligned}$$

But since we're integrating on $\partial B(0,\epsilon)$, we have $|y| = \epsilon$. The integral L_ϵ becomes:

$$\begin{aligned}
L_\epsilon &= -c_n(n-2)\epsilon^{1-n}\int_{\partial B(0,\epsilon)} f(x-y)\,dS(y) \\
&= -c_n(n-2)\epsilon^{1-n}\frac{n\alpha(n)\epsilon^{n-1}}{n\alpha(n)\epsilon^{n-1}}\int_{\partial B(0,\epsilon)} f(x-y)\,dS(y) \\
&= -c_n(n-2)\epsilon^{1-n} n\alpha(n)\epsilon^{n-1} \fint_{\partial B(0,\epsilon)} f(x-y)\,dS(y) \\
&= -\fint_{\partial B(0,\epsilon)} f(x-y)\,dS(y).
\end{aligned}$$

Thus $L_\epsilon \to -f(x)$ as $\epsilon \to 0$, establishing the theorem. ■

Definition 2.2.3. The equation:

$$-\Delta u = f$$

arising from Theorem 2.2.2 is called *Poisson's equation*.

II. Consequences of the Mean Value Property

Theorem 2.2.3 (Mean-Value Property). *Let $U \subset \mathbb{R}^n$ and $x \in U$. Suppose $u \in C^2(U)$ and $\Delta u = 0$ in U. For each $0 < r < \operatorname{dist}(x, \partial U)$, we have:*

$$u(x) = \fint_{B(x,r)} u(y)\,dy = \fint_{\partial B(x,r)} u\,dS.$$

Proof. Let $x \in U$. For each $0 < r < \mathrm{dist}(x, \partial U)$, define:

$$\begin{aligned}
\rho(r) &:= \fint_{\partial B(x,r)} u\, dS \\
&= \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} u\, dS \\
&= \frac{1}{n\alpha(n)r^{n-1}} \int_{S^{n-1}} u(x + r\omega) r^{n-1}\, d\omega \\
&= \frac{1}{n\alpha(n)} \int_{S^{n-1}} u(x + r\omega)\, d\omega
\end{aligned}$$

Taking the derivative of ρ with respect to r gives:

$$\begin{aligned}
\rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} \frac{d}{dr} u(x + r\omega)\, d\omega \\
&= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(x + r\omega) \cdot \omega\, d\omega.
\end{aligned}$$

Note that ω is precisely the outward normal on S^{n-1}. Making the substituion, $y = x + r\omega$ we have that $dy = r^{n-1} d\omega$ and $\omega = \frac{y-x}{r}$. Hence:

$$\begin{aligned}
\rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(x + r\omega) \cdot \omega\, d\omega \\
&= \frac{1}{n\alpha(n)r^{1-n}} \int_{\partial B(0,1)} Du(y) \cdot \frac{y - x}{r} dy.
\end{aligned}$$

The final integrand is by definition the outward normal derivative of $u(y)$. Green's identity yields:

$$\begin{aligned}
\rho'(r) &= \frac{1}{n\alpha(n)r^{1-n}} \int_{B(0,1)} \Delta u(y) dy \\
&= 0
\end{aligned}$$

Thus $\rho(r)$ is constant. With this fact, notice that.

$$\begin{aligned}
\rho(r) &= \lim_{r \to 0^+} \rho(r) \\
&= u(x).
\end{aligned}$$

This establishes $u(x) = \fint_{\partial B(x,r)} u\, dS$. Finally:

$$\begin{aligned}
\fint_{B(x,r)} u(y)dy &= \frac{1}{\alpha(n)r^n} \int_{B(x,r)} u(y)dy \\
&= \frac{1}{\alpha(n)r^n} \int_0^r \int_{S^{n-1}} u(x+\omega t)t^{n-1}\, d\omega dt \\
&= \frac{1}{\alpha(n)r^n} \int_0^r t^{n-1} \int_{S^{n-1}} u(x+\omega t)\, d\omega dt \\
&= \frac{1}{\alpha(n)r^n} \int_0^r n\alpha(n)t^{n-1} u(x) dt \\
&= u(x).
\end{aligned}$$

■

Theorem 2.2.4. *If $u \in C^2(U)$ satisfies $u(x) = \fint_{\partial B(x,r)} u dS$ for every $B(x,r) \subset U$, then $\Delta u = 0$.*

Proof. Suppose towards contradiction $\Delta u(x_0) > 0$ for some $x_0 \in U$. Then there exists some $\epsilon > 0$ such that $\Delta u(y) > 0$ for all $y \in B(x_0, \epsilon)$. For $0 < r < \epsilon$, define $\rho(r) := \fint_{\partial B(x_0,r)} u(y)dy$. By the same calculations made in Theorem 2.2.3, we have $\rho'(r) > 0$. But since $\rho(r) = u(x_0)$ by hypothesis, we also have $\rho'(r) = 0$. This is a contradiction, hence $\Delta u = 0$. ■

Theorem 2.2.5. *Let $U \subset \mathbb{R}^n$. Let $u \in C(U)$. If u satisfies:*

$$u(x) = \fint_{\partial B(x,r)} u dS = \fint_{B(x,r)} u(y)dy,$$

for any $B(x,r) \subset U$, then $u \in C^\infty(U)$.

Proof. Let $x \in U$. Choose $0 < \epsilon < \operatorname{dist}(x, \partial U)$. Then:

$$\begin{aligned}
u^\epsilon(x) &= (u * \eta_\epsilon)(x) \\
&= \int_{\mathbb{R}^n} u(x-y)\eta_\epsilon(y)\, dy.
\end{aligned}$$

Since η_ϵ is zero outside of $B(0,\epsilon)$, we have:

$$\begin{aligned}
u^\epsilon(x) &= \int_{B(0,\epsilon)} u(x-y)\eta_\epsilon(y)\, dy \\
&= \int_0^\epsilon \int_{S^{n-1}} u(x-r\omega)\eta_\epsilon(r\omega)r^{n-1}\, d\omega dr.
\end{aligned}$$

Since η_ϵ is a radial function, it does not depend on ω, allowing us to rearrange the above integral as follows:

$$u^\epsilon(x) = \int_0^\epsilon \eta_\epsilon(r\omega)r^{n-1} \left(\int_{S^{n-1}} u(x-r\omega)\, d\omega \right) dr.$$

By hypothesis, $u(x) = \fint_{\partial B(x,r)} u\,dS$ for any $B(x,r) \subset U$. Recalling that the surface area of the unit n-sphere is $n\alpha(n)$, we can rewrite the inner integral as:

$$\begin{aligned} u^\epsilon(x) &= \int_0^\epsilon \eta_\epsilon(r\omega) r^{n-1} \Big(n\alpha(n) u(x) \Big) dr \\ &= u(x) \int_0^\epsilon \eta_\epsilon(r\omega) r^{n-1} \Big(n\alpha(n) \Big) dr. \end{aligned}$$

But $n\alpha(n) = \int_{S^{n-1}} d\omega$. Substituting and rearranging gives:

$$\begin{aligned} u^\epsilon(x) &= u(x) \int_0^\epsilon \int_{S^{n-1}} \eta_\epsilon(r\omega) r^{n-1}\, d\omega dr \\ &= u(x) \int_{B(0,\epsilon)} \eta_\epsilon(y)\, dy \\ &= u(x). \end{aligned}$$

Thus $u^\epsilon(x) = u(x)$ on U_ϵ. By Theorem **??**, this establishes $u \in C^\infty(U_\epsilon)$. To conclude $u \in C^\infty(U)$, for any compact set $K \subset U$ and $\epsilon < \operatorname{dist}(K, \partial U)$, we have $K \subset U_\epsilon$, hence $u = e^\epsilon$ on K. Since K was arbitrary, we are done. ∎

Corollary 2.2.6. *Let $U \subset \mathbb{R}^n$. If u satsifies $\Delta u = 0$ in U, then $u \in C^\infty(U)$.*

Theorem 2.2.7 (Maximum Principle). *Let $U \subset \mathbb{R}^n$ and $u \in C^2(U) \cap C(\overline{U})$. Assume $\Delta u = 0$ in U. Then:*

(a) (Weak Maximum Principle) $\max_{\overline{U}} u = \max_{\partial U} u$;

(b) (Strong Maximum Principle) If U is connected, and there exists $x_0 \in U$ such that $u(x_0) = \max_{\overline{U}} u$, then u is constant on U.

Proof. (a) As $\partial U \subseteq \overline{U}$, we immediately have $\max_{\partial U} u \leqslant \max_{\overline{U}} u$. For the other direction of inequality, let $\epsilon > 0$. Consider the function $u + \epsilon|x|^2$. Then:

$$\begin{aligned} \Delta(u + \epsilon|x|^2) &= \Delta u + \epsilon \Delta |x|^2 \\ &= \epsilon \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial}{\partial x_j} |x|^2 \right) \\ &= \epsilon \sum_{j=1}^n \frac{\partial}{\partial x_j} 2x_j. \\ &= \epsilon \sum_{j=1}^n 2 \\ &= 2n\epsilon \\ &> 0. \end{aligned}$$

If $u + \epsilon|x|^2$ were to attain a local maximum at an interior point, then $\frac{\partial^2}{\partial x_j^2}(u + \epsilon|x|^2) \leqslant 0$ for all $j = 1, \dots, n$. This immediately implies that $\Delta(u + \epsilon|x|^2) \leqslant 0$, contradicting the above result.

Since the local maximum of $u + \epsilon|x|^2$ must lie on its boundary, we have:

$$\begin{aligned}\max_{\overline{U}} u &\leqslant \max_{\overline{U}}(u + \epsilon|x|^2) \\ &= \max_{\partial U}(u + \epsilon|x|^2)\end{aligned}$$

Taking $\epsilon \to 0$ gives $\max_{\overline{U}} u \leqslant \max_{\partial U} u$, establishing the claim.

(b) Assume $u(x_0) = \max_{\overline{U}} u := M$. Let $B(x_0, r) \subset U$. By the Mean Value Property:

$$\begin{aligned}M &= u(x_0) \\ &= ⨍_{B(x_0,r)} u(x)dx \\ &= \frac{1}{\alpha(n)r^n}\int_{B(x_0,r)} u(x)dx \\ &\leqslant \frac{M}{\alpha(n)r^n}\int_{B(x_0,r)} dx \\ &= M.\end{aligned}$$

If the maximum of $B(x_0, r)$ is $u(x_0)$, and if $⨍_{B(x_0,r)} u(x)dx = M$, it must be the case that $u(x) = M$ on all of $B(x_0, r)$. Since u is locally constant on a connected space, it is globally constant; i.e., $u(x) = M$ for all $x \in U$. ■

Remark. Note that the strong maximum principle implies the weak maximum principle.

Remark. A similar theorem can be stated in terms of local minima of functions, and the proof follows by consider $-u(x)$ and applying the maximum principle.

Corollary 2.2.8. *Let $y \in C(\partial U)$ and $f \in C(U)$. Solutions to the boundary value problem:*

$$\begin{cases}\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U\end{cases}$$

are unique.

Proof. Let u and v be solutions to the above boundary value problem. Since the Laplacian is linear, this induces the following boundary value problem:

$$\begin{cases}\Delta(u - v) = 0, & \text{in } U \\ u - v = 0, & \text{on } \partial U.\end{cases}$$

Since $u - v$ is harmonic in U, Theorem 2.2.7 and Theorem **??** assert that $\max_U(u - v) = \min_U(u - v)$. It must be the case that $u - v = 0$ everywhere on U, hence $u = v$. ■

Theorem 2.2.9 (Interior Derivate Estimate)**.** *Assume $\Delta u = 0$ in U. Then:*

$$|Du| \leqslant \frac{C}{r^{n+1}}\|u\|_{L^1(B(x_0,r))}$$

for each $B(x_0, r) \subset U$.

Proof. First, note that if u is a harmonic function then u_{x_i} is also harmonic (Clairaut's Theorem). As such, the function u_{x_i} admits the Mean-Value Property:

$$\begin{aligned}
|u_{x_i}(x_0)| &= \left| \fint_{B(x_0,\frac{r}{2})} u_{x_i}\,dx \right| \\
&= \frac{2^n}{\alpha(n)r^n} \left| \int_{B(x_0,\frac{r}{2})} u_{x_i}\,dx \right| \\
&= \frac{2^n}{\alpha(n)r^n} \left| \int_{\partial B(x_0,\frac{r}{2})} u\nu^i\,dS \right| \\
&\leqslant \frac{2^n}{\alpha(n)r^n} \sup\left\{|u(x)| : x \in \partial B\left(x_0, \frac{r}{2}\right)\right\} \int_{\partial B\left(x_0,\frac{r}{2}\right)} dS \\
&= \frac{2^n}{\alpha(n)r^n} \|u\|_{L^\infty(\partial B(x_0,\frac{r}{2}))}\, n\alpha(n)\frac{r^{n-1}}{2^{n-1}} \\
&= \frac{2n}{r} \|u\|_{L^\infty(\partial B\left(x_0,\frac{r}{2}\right))}
\end{aligned}$$

Note that for any $x \in \partial B(x_0, \frac{r}{2})$, we have $B(x, \frac{r}{2}) \subset B(x_0, r)$. Using this fact we can estimate $|u(x)|$ as follows:

$$\begin{aligned}
|u(x)| &= \left| \fint_{B(x,\frac{r}{2})} u(y)dy \right| \\
&= \frac{2^n}{\alpha(n)r^n} \int_{B(x,\frac{r}{2})} |u(y)|dy \\
&= \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x,\frac{r}{2}))} \\
&\leqslant \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0,r))}.
\end{aligned}$$

Taking the supremum over all $x \in \partial B(x_0, \frac{r}{2})$ gives:

$$\|u\|_{L^\infty(\partial B(x_0,\frac{r}{2}))} \leqslant \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0,r))}.$$

Combining this with our first string of inequalities results in:

$$\begin{aligned}
|u_{x_i}(x_0)| &\leqslant \frac{2n}{r} \cdot \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0,r))} \\
&= \frac{n2^{n+1}}{\alpha(n)r^{n+1}} \|u\|_{L^1(B(x_0,r))}.
\end{aligned}$$

Defining $C := \frac{n2^{n+1}}{\alpha(n)}$ and taking the gradient of $u_{x_i}(x_0)$ gives the desired result. ■

Theorem 2.2.10. *Assume $\Delta u = 0$ in U. Then:*

$$|D^\alpha u(x_0)| \leqslant \frac{C_k}{r^{n+|\alpha|}} \|u\|_{L^1(B(x_0,r))}$$

for each $B(x_0,r) \subset U$.

Proof. ∎

Corollary 2.2.11 (Liouville). *Suppose* $u : \mathbb{R}^n \to \mathbb{R}$ *is harmonic. If* u *is bounded, then* u *is constant.*

Proof. Let $x_0 \in \mathbb{R}^n$ and $r > 0$. Assume $|u(x)| < M$. For each $i = 1, \ldots, n$ we have:

$$\begin{aligned} |u_{x_i}(x_0)| &\leqslant \frac{C}{r^{n+1}} \|u\|_{L^1(B(x_0,r))} \\ &\leqslant \frac{C}{r^{n+1}} \int_{B(x_0,r)} M\,dx \\ &\leqslant \frac{CM}{r^{n+1}} \alpha(n) r^n \\ &= \frac{CM\alpha(n)}{r}. \end{aligned}$$

As we are assuming u is harmonic on all of $\mathbb{R}^n$, taking $r \to \infty$ does not fail any of the necessary assumptions made in Theorem 2.2.9. Since $u_{x_i}(x_0) = 0$, we conclude that u is constant. ∎

III. Green's Function, Poisson's Formula, and Analyticity

Example 2.2.2. Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1. Suppose $u \in C^2(U) \cap C^1(\overline{U})$. Can we write a formula for u in terms of its boundary values? Fix $x \in U$ and choose $\epsilon > 0$ such that $B(x,\epsilon) \subset U$. As a trick, consider Green's Formula III applied to the following integral:

$$\begin{aligned} &\int_{U\setminus B(x,\epsilon)} u(y)\Delta\Phi(y-x) - \Phi(y-x)\Delta u(y)\,dy \\ &= \int_{\partial(U\setminus B(x,\epsilon))} u(y)\frac{\partial\Phi}{\partial\nu}(y-x) - \Phi(y-x)\frac{\partial u}{\partial\nu}(y)\,dS(y) \\ &= \int_{\partial U} u(y)\frac{\partial\Phi}{\partial\nu}(y-x) - \Phi(y-x)\frac{\partial u}{\partial\nu}(y)\,dS(y) - \int_{\partial B(x,\epsilon)} \underbrace{u(y)\frac{\partial\Phi}{\partial\nu}(y-x)}_{(*)} - \underbrace{\Phi(y-x)\frac{\partial u}{\partial\nu}(y)}_{(**)}\,dS(y). \end{aligned}$$

First, note that $\Delta\Phi(x) = 0$, so our original integral can be simplified. Moreover, the derivations of $|K_\epsilon|$ and $|L_\epsilon|$ as $\epsilon \to 0$ from Theorem **??** can be replicated on (∗∗) and (∗) respectively. As such, we have:

$$\begin{aligned} \left|\int_{\partial B(x,\epsilon)} \Phi(y-x)\frac{\partial u}{\partial\nu}(y)\,dS(y)\right| &\to 0 \\ -\int_{\partial B(x,\epsilon)} u(y)\frac{\partial\Phi}{\partial\nu}(y-x) &\to u(x). \end{aligned}$$

All together, sending $\epsilon \to 0$ gives:

$$u(x) = -\int_U \Phi(y-x)\Delta u(y)\,dy - \int_{\partial U} u(y)\frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x)\frac{\partial u}{\partial \nu}(y)\,dS(y)$$

The first term of this integral is sticking around unless we assume u to be harmonic. The term $u(y)\frac{\partial\Phi}{\partial\nu}(y-x)$ is exactly what we were hoping for —it requires us to know information about u along its boundary. The term $\Phi(x-y)\frac{\partial u}{\partial \nu}$ is problematic however. With the given information, we are unable to compute $\frac{\partial u}{\partial \nu}(y)$. Our goal is to construct a *corrector function* which cancels with the unknown term.

Definition 2.2.4. Let $U \subset \mathbb{R}^n$. For $x, y \in U$ with $x \neq y$, Green's function for the region U is:

$$G(x,y) := \Phi(y-x) - \phi^x(y),$$

where $\phi^x(y)$ is a corrector function which solves the following boundary-value problem:

$$\begin{cases} \Delta\phi^x = 0, & \text{in } U \\ \phi^x = \Phi(y-x), & \text{on } \partial U \end{cases}.$$

Theorem 2.2.12. *If $u \in C^2(\overline{U})$ solves the boundary-value problem:*

$$\begin{cases} -\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases},$$

then:

$$u(x) = -\int_{\partial U} g(y)\frac{\partial G}{\partial \nu}(x,y)\,dS(y) + \int_U f(y)G(x,y)\,dy$$

Proof. ■

Theorem 2.2.13. *Suppose u solves the boundary-value problem:*

$$\begin{cases} \Delta u = 0, & \text{in } B^o(0,1) \\ u = g, & \text{on } \partial B(0,1) \end{cases}.$$

Then:

$$u(x) = \frac{1-|x|^2}{n\alpha(n)}\int_{\partial B(0,1)} \frac{g(y)}{|x-y|^n}\,dS(y).$$

Proof. We must determine the corrector function:

$$\begin{cases} \Delta\phi^x = 0, & \text{in } B^o(0,1) \\ \phi^x(y) = \Phi(y-x), & \text{on } \partial U \end{cases}.$$

Claim: $\phi^x(y) := \Phi\left(|x|(y-\tilde{x})\right)$ satisfies the above BVP, where $\tilde{x} := \frac{x}{|x|^2}$. Clearly $\phi^x(y)$ is harmonic for each $y \in B^o(0,1)$. Now let $y \in \partial B(0,1)$. ■

IV. Harnack's Inequality

Theorem 2.2.14 (Harnack's Inequality). *Let $U \subset \mathbb{R}^n$ be open and bounded. Assume $\Delta u = 0$ in U and $u(x) \geqslant 0$ for all $x \in U$. For any $V \subset\subset U$ which is open and connected, there exists a constant $C(U,V)$ depending only on U and V such that* $\sup_V u \leqslant C \inf_V u$.

Proof. Let $r = \frac{1}{4}\operatorname{dist}(V, \partial U)$. Let $x, y \in V$ with $|x-y| < r$. Then $B(x,r) \subset B(y,2r)$. By the Mean Value Principle:

$$\begin{aligned} u(x) &= ⨍_{B(x,r)} u(z)\,dz \\ &= \frac{1}{\alpha(n)r^n}\int_{B(x,r)} u(z)\,dz \\ &\leqslant \frac{1}{\alpha(n)r^n}\int_{B(y,2r)} u(z)\,dz \\ &= \frac{2^n}{\alpha(n)r^n 2^n}\int_{B(y,2r)} u(z)\,dz \\ &= 2^n u(y). \end{aligned}$$

Let $\{B_i\}_{i=1}^N$ be an open cover of $\overline{V}$ with finitely many balls of radius $\frac{r}{2}$. Since V is connected, for any $x, y \in V$ there is a chain $B_{i_1}, \ldots, B_{i_\ell}$ such that $x \in B_{i_1}$, $y \in B_{i_\ell}$, and $B_{i_j} \cap B_{i_{j+1}} \neq \emptyset$ for each $j = 1, \ldots, \ell - 1$. Pick an arbitrary $y_j \in B_{i_j} \cap B_{i_{j+1}}$ [1]. Then by our previous calculation:

$$\begin{aligned} u(x) &\leqslant 2^n u(y_1) \\ &\leqslant 2^n 2^n u(y_2) \\ &\vdots \\ &\leqslant 2^{\ell n} u(y). \end{aligned}$$

As $x \in V$ was arbitrary, we have $\sup_V u \leqslant C u(y)$. As $y \in V$ was arbitrary, we have $\sup_V u \leqslant C \inf_V u$. ∎

V. Energy Methods

Below is an alternative proof of uniqueness for the boundary value problem in Corollary 2.2.8 that does not rely on the Mean Value Property.

Lemma 2.2.15. *Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1. Suppose $u \in C^2(\overline{U})$. If:*

$$\begin{cases} \Delta u = 0, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases},$$

then $u \equiv 0$.

[1] We don't need to be concerned that the balls intersect only at a single point. As the open balls form a base for $\mathbb{R}^n$, we can find another open ball contained in each of their intersections.

Proof. We employ the following trick and then apply integration by parts:

$$\begin{aligned}
0 &= -\int_U u\Delta u\,dx \\
&= -\int_U \sum_{i=1}^n u u_{x_i x_i}\,dx \\
&= -\sum_{i=1}^n \int_U u u_{x_i x_i}\,dx \\
&= -\sum_{i=1}^n \left(-\int_U u_{x_i} u_{x_i}\,dx + \int_{\partial U} u u_{x_i} \nu^i\,dS\right) \\
&= \int_U |Du|^2\,dx.
\end{aligned}$$

Hence Du is zero, meaning that u is constant. Since u attains its maximum and minimum on its boundary, it must be the case that $u \equiv 0$. ■

Theorem 2.2.16. *Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1. Suppose $u \in C^2(\overline{U})$ satisfies:*

$$\begin{cases} -\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases}.$$

Then u is unique.

Proof. If u and v are solutions to the above boundary value problem, then $u-v$ satisfies the necessary conditions in Lemma 2.2.15. ■

Theorem 2.2.17 (Dirichlet Principle). *Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1. Consider the boundary value problem:*

$$\begin{cases} -\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases},$$

where $f \in C(U)$ and $g \in C(\partial U)$. Define:

$$\mathcal{A} := \{w \in C^2(\overline{U}) : w = g \text{ on } \partial U\}, \quad I[w] := \int_U \left(\frac{1}{2}|Dw|^2 - fw\right) dx$$

(a) *If $u \in \mathcal{A}$ satisfies the above BVP, then u minimzes I.*

(b) *If $u \in \mathcal{A}$ minimizes I, then u satisfies the above BVP.*

Proof. (a)Let $u \in C^2(\overline{U})$ satisfy the above BVP. Let $w \in \mathcal{A}$. Then:

$$\begin{aligned}
I[w] &= I[u+(w-u)] \\
&= \int_U \frac{1}{2}|D(u+(w-u))|^2 - f(u+(w-u))\,dx \\
&= \int_U \frac{1}{2}|Du+D(w-u)|^2 - fu - f(w-u)\,dx \\
&= \int_U \frac{1}{2}\left(|Du|^2 + 2Du\cdot D(w-u) + |D(w-u)|^2\right) - fu - f(w-u)\,dx \\
&= \int_U \frac{1}{2}|Du|^2 - fu\,dx + \int_U Du\cdot D(w-u) - f(w-u)\,dx + \int_U \frac{1}{2}|D(w-u)|^2\,dx \\
&= I[u] + \int_U Du\cdot D(w-u)\,dx - \int_U f(w-u)\,dx + \int_U \frac{1}{2}|D(w-u)|^2\,dx.
\end{aligned}$$

Applying Greens Formula II to $\int_U Du\cdot D(w-u)\,dx$ gives:

$$\int_U Du\cdot D(w-u)\,dx = -\int_U (w-u)\Delta u\,dx + \int_{\partial U} \nu\cdot D(w-u)u\,dS.$$

Note that $\Delta u = f$ in U and $w-u=0$ on ∂U. Hence:

$$\int_U (w-u)\Delta u\,dx = -\int_U (w-u)f\,dx.$$

The equation for $I[w]$ now simplifies to:

$$\begin{aligned}
I[w] &= I[u] + \int_U \frac{1}{2}|D(w-u)|^2\,dx \\
&\geqslant I[u].
\end{aligned}$$

Thus u minimizes I.

(b) Suppose $u \in \mathcal{A}$ minimizes I. Fix $v \in C_c^\infty(U)$. Define $i(t) := I[u+tv]$. Then:

$$\begin{aligned}
0 &= i'(0) \\
&= \int_U (Du\cdot Dv - fv)\,dx \\
&= \int_U (-v\Delta u - fv)\,dx \\
&= \int_U (-\Delta u - f)v\,dx
\end{aligned}$$

It must be the case that $-\Delta u = f$ in U. Hence U satisfies the above BVP. ∎

$\boxed{\S\ 2.3.}$